

Markscheme

Mathematics: analysis and approaches

Practice question bank

77. (a)

The population of a town is 12 000 at the start of year 1, and grows by 3.5% each year.

Find the first year in which the population exceeds 20 000.

model with $u_n = 12\,000(1.035)^{n-1}$; solve $12\,000(1.035)^{n-1} > 20\,000$ (M1)

$$(1.035)^{n-1} > 1.\overline{66} \quad \text{A1}$$

$$n - 1 > \frac{\ln(1.\overline{66})}{\ln(1.035)} = 14.8\dots \quad \text{(M1)}$$

testing integers confirms the population first exceeds 20 000 in year 16 A1

[5 marks]